

II-Probabilistic Approaches Probability, Marginal and Conditional Distributions

Foundations of Neuro-Symbolic AI

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Outline

- ▶ **Uncertainties**
- ▶ Probability Distributions
- ▶ Marginal Distributions
- ▶ Conditional Distributions
- ▶ Bayes Theorem

Toy accounting example

Let us imagine that each possible assignment to is of equal frequency.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} [X_{A1}, X_{A2}, X_F]$$

We can ask the following questions: Is the account **Account 1** or the account **Account 2** more likely to be booked?

Since logic can only handle certainty, it cannot answer this question.

However, we can intuitively assign probabilities to the states:

$$\begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \frac{1}{3} & 0 \end{pmatrix} [X_{A1}, X_{A2}, X_F]$$

There are two worlds where **Account 1** is booked, and one world where **Account 2** is booked, thus

$$\mathbb{P}[X_{A1} = 1] = \frac{2}{3} > \mathbb{P}[X_{A2} = 1] = \frac{1}{3}.$$

Uncertainty: Summarizing the unknown

Uncertainties need to be handled:

- ▶ Not all variables are measured.

Example: When we do not know the value of **Feature**, then we do not know whether **Account 1** or **Account 2** have to be booked.

- ▶ Parts of the models are not known.

Example: Even if **Feature** is known to be **False**, then we do not know whether **Account 1** or **Account 2** have to be booked.

Logics cannot handle uncertainties.

But: They are already built in model counts.

Comparing logics and probability theory

Modelling assumptions

- ▶ **Ontological:** What properties does a system have?
→ We assume **fixed sets of categorical properties**.
- ▶ **Epistemological:** What can we tell about these properties?
→ We associate **possibilities or probabilities** with states.

Two main frameworks differing in the **empistemological** assumptions:

- ▶ Logics (**Possibilities**): The traditional mathematical model of intelligence.
- ▶ Probability Theory (**Probabilities**): Handling uncertainties about the systems state.

In both cases, we encode our knowledge in an arrangement of numbers:
Tensors with efficient representations by **Tensor Networks**.

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Recap: Tensors as Propositional Formulas

So far: Tensors with Boolean coordinates and leg dimension 2
Generalize to

- ▶ arbitrary leg dimensions
- ▶ real valued coordinates instead of Booleans $\{0, 1\}$

Probability distributions

Instead of storing 0/1 for possibilities, we can store probabilities (in the interval $[0, 1]$) in the coordinates of a tensor.

The total probability of all states must be 1 and can be calculated by contracting all legs of the tensor since

$$\sum_{\mathbf{x}_{[d]}} \mathbb{P} [X_{[d]} = \mathbf{x}_{[d]}] = \langle \mathbb{P} [X_{[d]}] \rangle_{[\emptyset]} .$$

Definition (Probability Tensor)

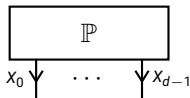
Let there be a factored system $X_{[d]}$ defined by variables X_k taking values in $[m_k]$. A probability distribution over the states of $X_{[d]}$ is a tensor $\mathbb{P} [X_{[d]}]$ with coordinates in the interval $[0, 1]$ such that

$$\langle \mathbb{P} [X_{[d]}] \rangle_{[\emptyset]} = 1 .$$

Normalization constraint by directionality

Probability distributions sum to 1.

We depict this by outgoing legs:



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Example: Being at a dentist

We are reasoning about a factored system with three variables:

- ▶ **Toothache** $x_0 \in [2]$, whether your tooth hurts
- ▶ **Cavity** $x_1 \in [2]$, whether there is a cavity in your tooth
- ▶ **Catch** $x_2 \in [2]$, whether the dentist catches in your tooth

Example: Being at a dentist

Let us take the following probability distribution over the states of the system:

Toothache	x_0	Cavity	x_1	Catch	x_2	$\mathbb{P}[X_0 = x_0, X_1 = x_1, X_2 = x_2]$
0		0		0		0.576
0		0		1		0.144
0		1		0		0.008
0		1		1		0.072
1		0		0		0.064
1		0		1		0.016
1		1		0		0.012
1		1		1		0.108

We align the probability values in the coordinates of an order-3 tensor:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

Marginal Distributions

Example: What is the probability that there is a cavity (or not) when having toothache (or not)?

- ▶ The variable **Catch** is not relevant for this question, thus we want to **marginalize** over it.
- ▶ To this end, we sum the probability of states differing in the value of **Catch** only.

This amounts to a contraction of the probability distribution tensor closing the variable X_2 :

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

↓

$$\begin{pmatrix} * & * \\ * & * \end{pmatrix} [X_0, X_1]$$

Marginal Distributions

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$$\begin{pmatrix} 0.72 & * \\ * & * \end{pmatrix} [X_0, X_1]$$

Marginal Distributions

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$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1]$$

Marginal Distributions

Summation of probabilities of states are done by contractions of the probability distribution tensor.

Definition (Marginal Distribution)

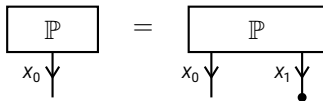
Given a probability distribution $\mathbb{P} [X_{[d]}]$ and a subset $\mathcal{U} \subset [d]$, the marginal distribution of the variables $X_{\mathcal{U}}$ is the distribution

$$\mathbb{P} [X_{\mathcal{U}}] = \langle \mathbb{P} [X_{[d]}] \rangle_{[X_{\mathcal{U}}]} .$$

Marginal Distributions in Contraction Formalism

Contractions are useful to

- Specify the Probability Tensor, or a decomposition of it
- Specify the variables to marginalize over as the ones left open



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Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} * & * \\ * & * \end{pmatrix} [X_0|X_1]$$

This is called a **conditional** distribution.

Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} \frac{0.72}{0.72+0.08} & \frac{0.08}{0.72+0.08} \\ * & * \end{pmatrix} [X_0|X_1]$$

This is called a **conditional** distribution.

Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix} [X_0 | X_1]$$

This is called a **conditional** distribution.

Normations

More general, conditionalization amounts to normalization of probability tensors.

Definition (Normalization)

The **normalization** of a tensor $\tau [X_{[d]}]$ with respect to disjoint subsets $\mathcal{V}^{\text{out}}, \mathcal{V}^{\text{in}} \subset [d]$ is the tensor

$$\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}]}$$

defined by the coordinates

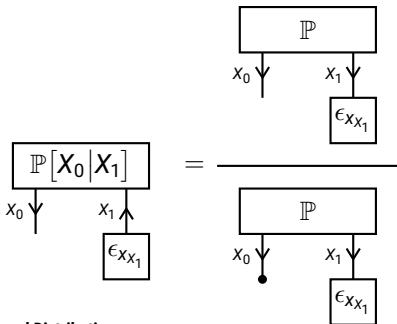
$$\begin{aligned} & \langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} = x_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}] } \\ &= \begin{cases} \frac{\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} = x_{\mathcal{V}^{\text{out}}}, X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]}}{\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]}} & \text{if } \langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]} \neq 0 \\ \frac{1}{\prod_{v \in \mathcal{V}^{\text{out}}} m_v} & \text{else} \end{cases} \end{aligned}$$

Formal Definition of Conditional Distributions

Definition (Conditional Probability)

Given a distribution $\mathbb{P}[X_{[d]}]$ and disjoint subsets $\mathcal{V}^{\text{out}}, \mathcal{V}^{\text{in}} \subset [d]$, the distribution of $X_{\mathcal{V}^{\text{out}}}$ conditioned on $X_{\mathcal{V}^{\text{in}}}$ is the normalization

$$\mathbb{P}[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}] = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}]}$$



Directed notation

Directed tensors represent normations:

Each slice of the incoming variables is a normalized tensor, i.e. coordinates summing to 1.

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The Bayes Theorem

The Bayes Theorem relates conditional probabilities:

Theorem (Bayes Theorem)

For any joint distribution of two categorical variables X_0 and X_1 it holds that

$$\mathbb{P}[X_0|X_1] = \frac{\mathbb{P}^{X_0, X_1}}{\mathbb{P}^{X_1}} = \frac{\mathbb{P}[X_1|X_0] \mathbb{P}^{X_0}}{\mathbb{P}^{X_1}}.$$

Bayes Theorem in the Dentist Example

Directions of Reasoning

- ▶ **Causal direction:** Toothache is caused by cavity
- ▶ **Diagnostic direction:** Cavity is probable because of toothache

Bayes Theorem allows us to reason in diagnostic direction, given an underlying causal influence:

$$\begin{aligned}\mathbb{P}[\text{Cavity}|\text{Toothache}] \\ = \mathbb{P}[\text{Toothache}|\text{Cavity}] \frac{\mathbb{P}[\text{Cavity}]}{\mathbb{P}[\text{Toothache}]}\end{aligned}$$

Contraction Calculus for Probability Tensors

Probabilistic queries can be answered by contractions

- Marginal probabilities

$$\mathbb{P}^{X_0} = \langle \mathbb{P} \rangle_{[X_0]}$$

- Conditional probabilities

$$\mathbb{P}[X_0|X_1] = \langle \mathbb{P} \rangle_{[X_0|X_1]}$$

Outlook: **Tensor network decompositions of $\mathbb{P}$** increase the execution efficiency!

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Probabilistic Queries

The probability of a formula f to hold

$$\sum_{x_{[d]} : f[X_{[d]}=x_{[d]}]=1} \mathbb{P} [X_{[d]} = x_{[d]}] = \langle \mathbb{P} [X_{[d]}] , f [X_{[d]}] \rangle_{[\emptyset]}$$

Generalization of entailment: A formula is entailed if the probability is 1.